Template: Challenge Exploration

List the 10 most relevant or important questions and areas of interest,
List 10 associated stakeholders and other relevant sources of information.

AREAS OF INTEREST / QUESTIONS

(Write down 10 Areas of interest, that can be connected to more than one Stakeholder.)

1. **Economics –** What are the expected implementation and operating costs? What funding opportunities are available?
2. **Implementation timeline –** Duration, what are the key milestones, how long for each goal and communication plan troughout the project?
3. **Social Change** – How will it affect the stakeholders, during and after the implementation.
4. **Work Involvement –** What training, skills and type of background is needed for the implementation.
5. **Environmental impact –**How will the solution contribute to sustainability and reduce environmental impacts?
6. **Mental Health –** Can the changes affect anyone mentally, cause stress etc.
7. **Safety and Secrurity –** What technical and public safety risks needs to be adressed
8. **Physical Health –** Are there any potential physical risks when it comes to the project
9. **Comunity Development –** What is the end goal of the project, how can it bring value in the future
10. **Ethics –** How can we ensure fairness for everyone and transparency regarding the project

STAKEHOLDERS

(Write down 10 stakeholders, that can be connected to more than one question/area of interest)

- Employees (of any kind) at BEOF
- Residents of Bornholm
- Rønne Vand og Varme (RVV)
- Municipality of Bornholm
- Danish Government
- Energistyrelsen
- Private owners of Windmills and Solar Plants
- Shop/business Owners on Bornholm
- European Social Funds Plus
- Researchers in this Field

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Stakeholder 1

Who is the Stakeholder? **What** is their role in regard to the challenge?

Residents of Bornholm are the end users and one of the key stakeholders in the transition to a more electrified and renewable local heating system. They are directly affected by changes in the district heating system, including heating costs, reliability, infrastructure development and the environmental impacts of the transition. Their perspectives are therefore essential for ensuring public acceptance and developing a solution that meets community needs. Residents stand to benefit from long term price stability and a cleaner environment. However, they may fear short term heating price spikes or the visual impact of new infrastructure on their local landscape.

How will you collect information? (e.g., interview, written questions, observations on location, etc.) and why have you chosen these?

We will use digital surveys, short interviews with local residents, and existing reports from BEOF and Bornholm Municipality. Surveys allow us to reach a broad range of residents, while interviews provide deeper insight into their experiences and concerns. Existing research helps us understand the local context and validate our findings.

How will you establish contact? (e.g., invite for a MS Teams meeting, phone call, etc.) **What** is your plan B?

Our primary contact channels will be local facebook community groups and if available, BEOF customer communication channels. If the response rate is lower than expected we may supplement the data collection by face to face interviews respectfully in public areas during our field visit to Bornholm.

Who in your team will contact the stakeholder & **When**?

Kani will be responsible for initiating contact with the residents and the overalll coordination of communication. Perhaps using their mail, so all emails and messages are saved in one place. During the field trip, two members (Jiayue and Melek) will conduct the interviews, while another team member will document the findings. Either by writing it down or perhaps recording if allowed by the residents. If more residents are being interviewed at the same time, the remaining team members can conduct that interview.

How will you document the data? (e.g., audio recording, notes, photos)

All written communication (emails, survey responses,and messages) will be securely stored for documentation purposes. During interviews, notes will be taken, and if the participant consent, audio recordings may be used to ensure accurate documentation.

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Stakeholder 2:

BEOF: District Heating / Project Development employee

Who is the Stakeholder? **What** is their role within the challenge?

An employee at BEOF working with district heating, project development or technical planning. BEOF operates the district-heating system in Nexø and is therefore a key decision-maker in the transition toward electrification. This stakeholder can provide insight into technical feasibility, investment priorities, security of supply, customer needs and practical implementation challenges.

They may view electrification positively because it can reduce biomass dependency and increase flexibility, but may also be concerned about investment costs, grid capacity, reliability and implementation risks.

How will you collect information? (e.g., interview, written questions, observations on location, etc.)

We will conduct a 30–45 minute semi-structured interview. An interview is suitable because we need detailed information about technical constraints, priorities and trade-offs that cannot easily be captured through a questionnaire. Semi-structured questions also allow us to ask follow-up questions when new issues arise.

How will you establish contact? (e.g., invite for a MS Teams meeting, phone call, etc.) **What** is your plan B?

Monday morning when travelling to Bornholm, we will ask Dorthe Skovgaard Lund for a contact at BEOF, then contact them by email and request an interview with an available project development employee involved in district heating or project development. If we do not receive a response during Monday, we will follow up by phone.

Plan B: Ask our DTU facilitator or BEOF contact person to connect us with another relevant employee. If an in-person meeting is not possible, we will conduct the interview through Teams or phone.

Who in your team will contact the stakeholder & **When**?

Else Marie will be responsible for contacting BEOF. Initial email will as mentioned be sent Monday morning. Follow-up; as we only have a few days on Bornholm, we will have to follow up by Monday afternoon and then by Tuesday resort to plan B. Interview target: during the Bornholm visit or immediately after we arrive homer over teams. The last two days of week two should preferably be kept as a buffer for rescheduling or follow-up questions.

How will you document the data? (e.g., audio recording, notes, photos)

With the stakeholder's permission, we will audio-record the interview and take written notes. After the interview, we will summarize the main findings under themes such as technical feasibility, cost, resilience, sustainability and community impact. Relevant quotes and observations will be documented anonymously unless permission is given to identify the interviewee.

Template: Challenge Exploration

Stakeholder 3

Who is the Stakeholder? **What** is their role within the challenge?

The Regional Municipality of Bornholm, the owner of BEOF, secures energy for approx. 40,000 residents. As the local planning authority, it is the key gatekeeper for new infrastructure and leads public consultations. While this transition helps the municipality meet climate goals and attract green investment, it also introduces significant political risk regarding public resistance to land use changes.

How will you collect information? (e.g., interview, written questions, observations on location, etc.)

We will contact relevant people from the municipality, particularly those involved in the Energiø Bornholm and Energipark projects, to request interviews, using a prepared list of questions sent by email in advance. During our trip, we will enrich these interviews with direct observations. Before and after the trip, we will enrich our understanding with publicly available information, such as the municipality's official website and other public documents.

How will you establish contact? (e.g., invite for a MS Teams meeting, phone call, etc.) **What** is your plan B?

We will use our challenge partners existing connections to obtain contact information (email or phone) for the people managing the projects mentioned above. Once we have this, we will send them our prepared questions and request an interview, which we expect to work given the challenge partners existing relationship with the municipality. If this is unsuccessful, our fallback is to contact the municipalities official press contact or relevant department directly to request an interview or information.

Who in your team will contact the stakeholder & **When**?

Nikolas and Prithvi will start asking for contact information on Monday when we arrive on Bornholm. If we don't have useful contact information by Wednesday, we will send our questions to the press contact by Thursday at the latest. Giving us approximately one week to receive answers.

How will you document the data? (e.g., audio recording, notes, photos)

We will share contact information in our team chat and create a shared document in our Teams group, so everyone has access in case of illness. For emails sent to the municipality, all group members will be added in the CC so everyone can follow the conversations. For interviews, we will ask for permission to record, and upload the recordings afterward to the shared Teams folder.